Machine Condition Monitoring — Oct 14th, 2026

1. Moment of Inertia for I-beam.

Weight Range:
100 Kg
↓
400 Kg

E: 200×10^9

$I_z = 1.75659 \times 10^9 \text{ mm}^4$

↓

$1.75659 \times 10^{-3} \text{ m}^4$

$EI = 200 \times 10^9 \cdot 1.75659 \times 10^{-3}$

$= 3.5132 \times 10^8 \text{ Nm}^2$

2. Calculate the beam stiffness (for both Configuration)

$L = 4m$

$K = F/\delta$

↓

$3LEI/(a^3b^2)$

Case	a	b	k (N/m)
1	2	2	$3(4)(3.5132\times10^8)/(2^2\cdot2^2) = 2.635\times10^8$
2	3	1	$3(4)(3.5132\times10^8)/(3^2\cdot1^2) = 4.684\times10^8$

3. Natural Frequency of system in Hz (for both Configuration)

$W_n = \sqrt{(K/m)}$ $f_n = W_n/2\pi$

Case	m (Kg)	W_n (rad/s)	f_n (Hz)
1 (a = 2m)	100	$\sqrt{(2.635\times10^8/100)} = 1623.3$	$1623.3/2\pi = 258.4$
1 (a = 2m)	400	$\sqrt{(2.635\times10^8/400)} = 811.6$	$811.6/2\pi = 129.2$
2 (a = 3m)	100	$\sqrt{(4.684\times10^8/100)} = 2164.3$	$2164.3/2\pi = 344.5$
2 (a = 3m)	400	$\sqrt{(4.684\times10^8/400)} = 1082.1$	$1082.1/2\pi = 172.2$

4. Critical Damping

$C_c = 2\sqrt{Km} = 2m\omega_n$ (Assumed $\zeta = 0.15$)

Case	m (Kg)	C_c (N·s/m)	$C = \zeta \cdot C_c$
1	100	$2(100)(1623.3) = 324660$	$0.15 \cdot 324660 = 48699$
1	400	$2(400)(811.6) = 649280$	$0.15 \cdot 649280 = 97392$
2	100	$2(100)(2164.3) = 432860$	$0.15 \cdot 432860 = 64929$
2	400	$2(400)(1081.1) = 864880$	$0.15 \cdot 864880 = 129732$

5. Damped natural Frequency

$\omega_d = \omega_n\sqrt{(1-\zeta^2)}$ $\quad f_d = \omega_d / 2\pi$

Case	m (Kg)	ω_d (rad/s)	f_d (Hz)
1	100	$(1623.3)\sqrt{1-0.15^2} = 1604.9$	$1604.9/2\pi = 255.4$
1	400	$(811.6)\sqrt{1-0.15^2} = 802.4$	$802.4/2\pi = 127.7$
2	100	$(2164.3)\sqrt{1-0.15^2} = 2139.8$	$2139.8/2\pi = 340.6$
2	400	$(1081.1)\sqrt{1-0.15^2} = 1068.9$	$1068.9/2\pi = 170.1$